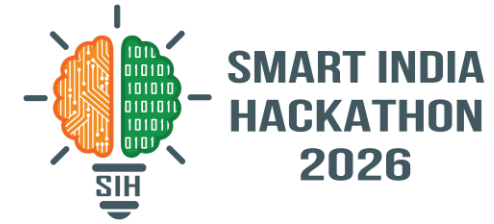
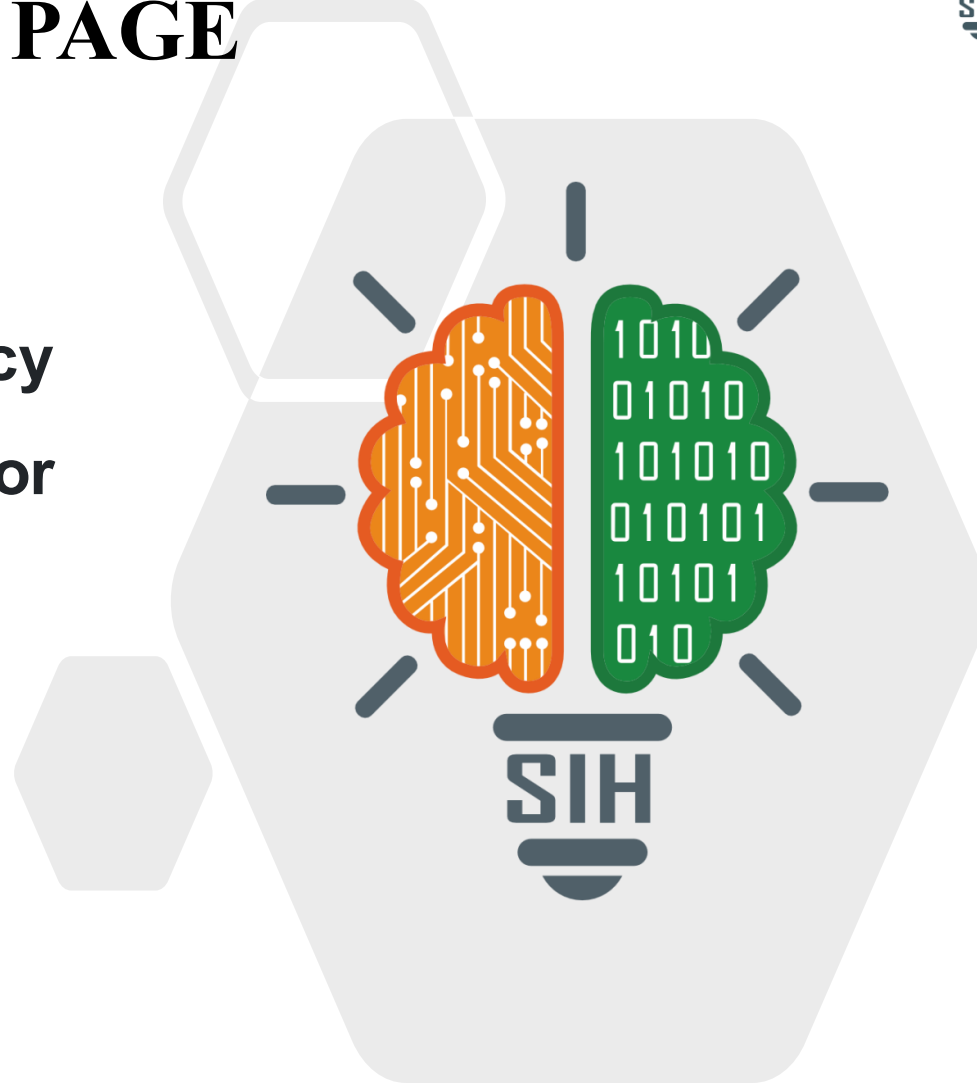


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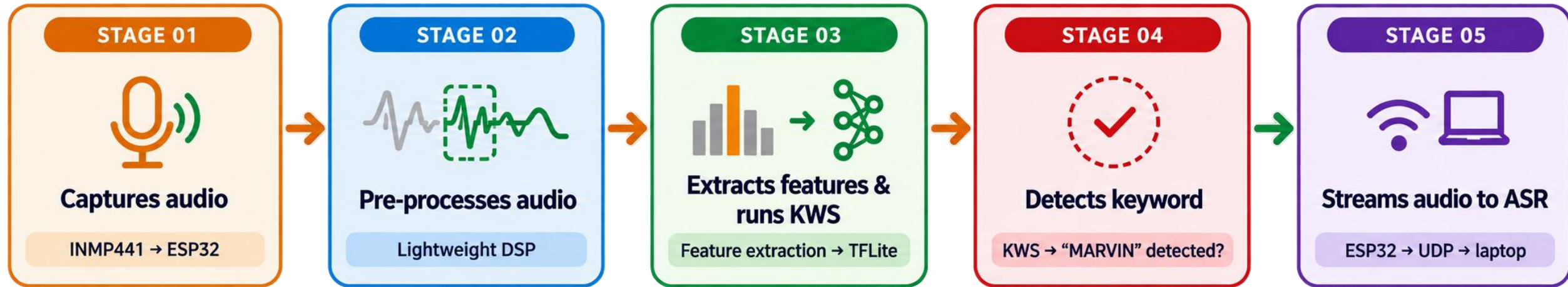
TITLE PAGE



- Problem Statement ID – SIH26172
- Problem Statement Title- Low Latency and Efficient Voice Activator for Edge Devices
- Theme- Smart Automation
- PS Category- Hardware
- Team Name :- MockingJays



IDEA TITLE



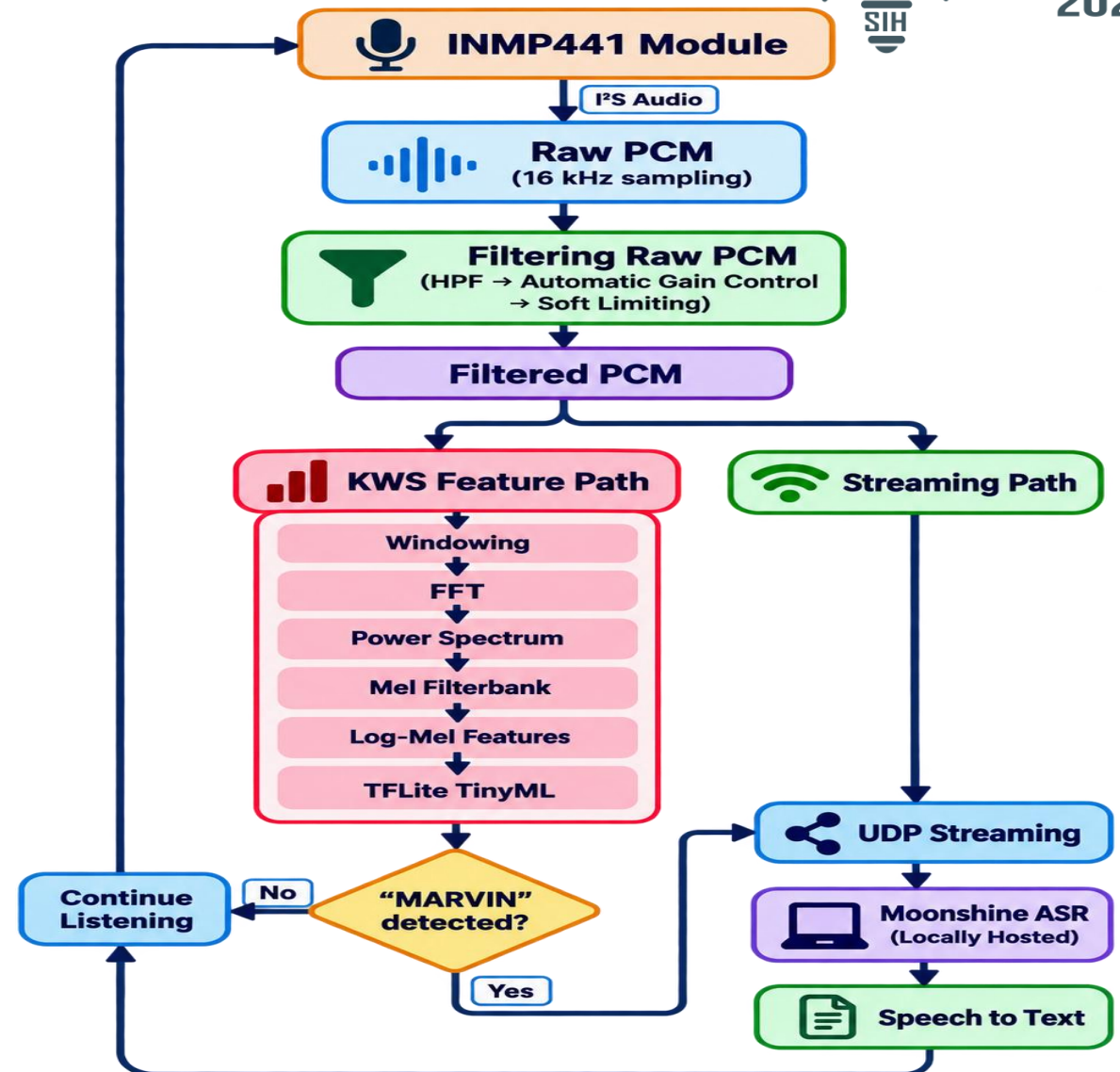
- Captures raw microphone audio from inmp441 module
- Sampling data at 16 kHz PCM
- Divide it into 256 chunks
- Put it into Ring buffer for continuous audio sampling
- High-pass filtering to remove noise
- Automatic gain control and Soft limiting to smoothen the data
- Optimized to reduce unnecessary CPU processing
- Windowing and FFT converts raw data to power spectrum.
- Through Mel filterbank store Log-Mel features
- Feed it to TFLite for keyword spotting mode
- Continuously monitors the processed audio
- If "MARVIN" is not detected → continue listening
- If "MARVIN" is detected → activate transmission
- Sends filtered PCM audio after activation.
- Low-latency UDP transmission
- Moonshine AI serves as our ASR server
- Output: recognized text / command

1) Technologies Used:-

- **Hardware:** ESP32 • INMP441 I²S Microphone
- **Firmware:** C • ESP-IDF • FreeRTOS
- **Edge AI:** TensorFlow Lite • Keyword Spotting
- **Signal Processing:** HPF • AGC • Soft Limiter • FFT • Log-Mel Features
- **Communication & ASR:** Wi-Fi • UDP • Moonshine AI (Local ASR)

2) Key Technical Approach:-

- **Always-On Edge Processing:** Continuous 16 kHz audio acquisition and lightweight preprocessing on ESP32
- **TinyML Wake-Word Detection:** Log-Mel audio features are processed locally by the TFLite KWS model to detect “MARVIN.”
- **Conditional ASR Streaming:** Filtered PCM is transmitted via UDP to locally hosted Moonshine ASR only after wake-word activation.



ASR Model Comparison Vosk vs Whisper Base vs Moonshine

Parameter	Vosk (Small)	Whisper (Base)	Moonshine (Base)
CPU Usage	~100% of 1 core	317 - 437% (multi-core)	Optimized, edge-efficient
Latency	~1.29 s	~0.86 s (RTF 0.237)	Ultra-low, 5-15× faster
Accuracy	Poor in initial tests	60% clean, 0% noisy	10.07% WER, competitive
Real-time Performance	Streaming supported, slow chunks	Weak streaming, RTF 5.2	Fast real-time, edge-ready

Based on our testing, Moonshine provides the smallest footprint, fully offline operation with zero cloud costs, and the best latency-to-accuracy balance for real-time edge voice apps.

```

CPU: 11.42% | RAM Used: 40.65% | RAM Free: 59.35%
CPU: 11.83% | RAM Used: 40.65% | RAM Free: 59.35%
CPU: 11.63% | RAM Used: 40.89% | RAM Free: 59.11%
CPU: 11.75% | RAM Used: 40.65% | RAM Free: 59.35%

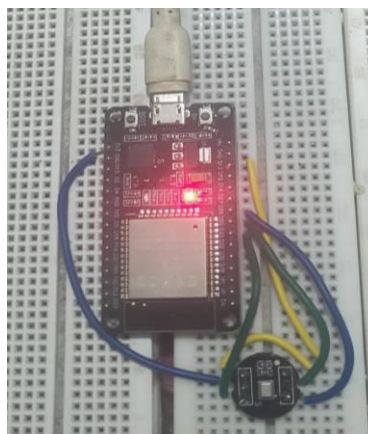
Received 512 bytes
Stopping...

Final transcription:
[0.70s] Hey Marvin what's the weather today?
UDP socket closed.

C:\Users\PRATIKA\vosk_test>

```

We reported only 10% CPU utilization for UDP data transmission to the ASR server with 95% accuracy



Circuit Connection during testing

- The prototype proves technical feasibility with low resource use, while offline operation, low hardware costs, and reduced cloud dependency make it ideal for scalable edge deployment.

Challenges faced and planned solution:-

- PROCESSING LATENCY** → Minimal on-device DSP for KWS; offload additional filtering to ASR host.
- FALSE TRIGGERS** → ~95% confidence threshold to reject low-confidence wake-word detections.
- LIMITED MEMORY** → 256-sample chunk processing with ring-buffer memory reuse.
- ASR LATENCY** → Use of moonshine ai for ASR server which provided the least latency and accuracy for speech processing.

1) RESPONSIVE:- Low-Latency Voice Activation

- Real-time on-device wake-word detection
- Eliminates cloud round-trip for activation

2) PRIVATE:- Local-First Voice Processing

- Continuous audio processed locally
- Audio streamed only after wake-word trigger

3) ECONOMICAL:- Reduced Cloud Dependency

- No continuous cloud ASR/API usage
- Lower bandwidth & operational overhead

4) DEPLOYABLE:- Scalable Edge-AI Architecture

- Works with local/offline infrastructure
- Adaptable to IoT, assistive & industrial devices



- **Audio Playback Without Libraries: From Raw PCM to Sound on Falstad and ESP32 by Iftimie Alexandru**
- **FreeRTOS (Supplemental Features) for ringbuffers by espidf**
- **github.com/Shivp1413/I2S-microphone-live-stream:- referred for audio processing and filtering noisy data.**
- **Moonshine AI, Moonshine Voice: Open-Source Voice Toolkit, GitHub.
<https://github.com/moonshine-ai/moonshine>**